

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1562

Fort Loudon Dam 138 kV
September 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1562. The Interconnection Customer proposes a 114.5 MW solar photovoltaic facility interconnecting at the Fort Loudon Dam 138 kV substation in the Virginia area. The request was received on August 31, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 574 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,131,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Big Hanaford – Carver 138 KV	Plaza – Big Hanaford 138 KV	162.9	264	24.6
Fritch – River Rd 138 KV	Fritch – Lakeside 138 KV	151.7	208	21.0
Shockoe – Carver 138 KV	Plaza – Big Anaford 138 KV	145.3	344	24.6
Turner Rd – Elko 138 KV line	Mitchell Dam – Haggin 345 KV	142.8	315	10.4
Shack Mills Sta – Kimberley 138 KV L	Bear Island – Chancellor Row 345 KV L	137.8	249	7.7
Riv Rd – Humacao 2601 138 KV	Fritch – Lakeside 138 KV	131.5	224	21.0
Chancellor Row – Shack Mills Sta 138 KV L	Chancellorrow – Bear Island 345 KV	130.9	284	7.7
Paragould Reciprocating – Kevlar 138 KV L	Plaza – Big Hanaford 138 KV line	126.6	237	12.9
Manchester Substation – Shockoe 138 KV	Plaza – Big Hanaford 138 KV	124.9	306	11.6
Dps Ocean Spray – Four Rivers 138 KV	Leavitt – Birchwood 345 KV line	123.0	329	5.5
Turner Rd – Tyler 138 KV line	Mitchelldam – Haggin 345 KV	122.2	392	10.4
King City – Centralia 138 KV L	Big Hanaford – Fritch 138 KV	121.4	353	51.0
Moonstone – Paragould Reciprocating 138 KV	Big Hanaford – Plaza 138 KV	120.1	217	12.9
Trenton Diesel – Four Rivers 138 KV	Leavitt – Birchwood 345 KV L	118.0	343	5.5
Kimberley – Elmont 138 KV Line	Bear Island Substation – Chancellor Row 345 KV	117.2	273	7.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Northern Neck 345-138 kV Transformer Bank	Dunn Siding – Bear Island 345 KV	106.9	496	5.1
Toro Negro 2 9901 – Anoka 138 KV line	Centralia – King City 138 KV	103.7	196	28.6
Strathmore – Humacao 2601 138 KV	Fritch – Lakeside 138 KV	103.0	256	21.0
Deltaport – Moonstone 138 KV line	Fritch – Big Hanaford 138 KV Line	102.8	298	32.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carver – Big Hanaford 138 KV	Rebuild line with higher-capacity conductor	\$1,248,000	9% (\$110,000)
Fritch – River Rd 138 KV	Rebuild line with higher-capacity conductor	\$3,371,000	9% (\$315,000)
Carver – Shockoe 138 KV	Rebuild line with higher-capacity conductor	\$844,000	9% (\$74,000)
Turner Rd – Elko 138 KV line	Rebuild line with higher-capacity conductor	\$4,123,000	5% (\$198,000)
Shack Mills Sta – Kimberley 138 KV line	Rebuild line with higher-capacity conductor	\$2,673,000	4% (\$116,000)
Riv Rd – Humacao 2601 138 KV	Rebuild line with higher-capacity conductor	\$2,015,000	9% (\$188,000)
Chancellor Row – Shack Mills Station 138 KV	Rebuild line with higher-capacity conductor	\$2,230,000	4% (\$96,000)
Paragould Reciprocating – Kevlar Substation 138 KV	Rebuild line with higher-capacity conductor	\$2,019,000	9% (\$181,000)
Shockoe – Manchester Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,098,000	9% (\$354,000)
Dps Ocean Spray – Four Rivers 138 KV	Rebuild line with higher-capacity conductor	\$6,586,000	5% (\$327,000)
Tyler – Turner Road 138 KV	Rebuild line with higher-capacity conductor	\$8,007,000	5% (\$385,000)
King City – Centralia 138 KV L	Rebuild line with higher-capacity conductor	\$7,206,000	17% (\$1,218,000)
Moonstone – Paragould Reciprocating 138 KV	Rebuild line with higher-capacity conductor	\$2,335,000	9% (\$210,000)
Trenton Diesel – Four Rivers 138 KV	Rebuild line with higher-capacity conductor	\$9,021,000	5% (\$448,000)
Kimberley – Elmont 138 KV Line	Rebuild line with higher-capacity conductor	\$1,552,000	4% (\$67,000)
Northern Neck 345-138 kV XFMR	Replace transformer with higher-rated unit	\$15,586,000	3% (\$434,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Toro Negro 2 9901 – Anoka 138 KV	Reconductor line	\$841,000	27% (\$230,000)
Strathmore – Humacao 2601 138 KV	Reconductor line	\$1,008,000	9% (\$94,000)
Deltaport – Moonstone 138 KV	Reconductor line	\$736,000	12% (\$86,000)
	Total allocated to project		\$5,131,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.